TECHNICAL PROJECT PORTFOLIO

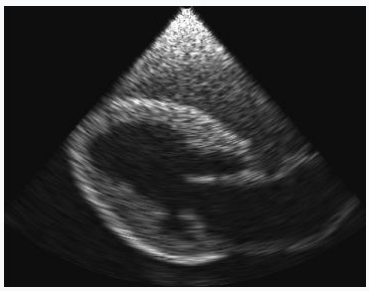
Stefan Klemmer Chandía

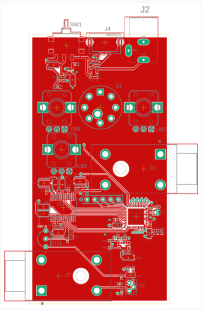
Embedded Electronics · Measurement Systems · Simulation · Signal Processing

Selected projects demonstrating complete development cycles: from physical modelling and circuit design to PCB layout, embedded software and prototyping, through to numerical simulation, signal processing and validation.

01	**Mechanical Ventilator Prototype**	Physical Modelling · Parameter Estimation · Optimal Control (LQR) · PCB · C/Raspberry Pi
02	**Dynamic Cardiac Elastography Simulation**	CT-Based Geometry · Devito/FDM · Field II · Signal Processing
03	**Auto-Wah Guitar Pedal**	Circuit Design · Simulation · PCB Design · Design Review







COMPETENCIES ACROSS THE THREE PROJECTS

Hardware	Embedded systems	Simulation	User interface	V&V
Circuit design PCB EAGLE · EasyEDA	C/C++ Microcontroller · Pi FPGA	LTSpice · Devito/FDM MATLAB · Field II	Typescript · CSS · GTK	Measurement · Analysis Documentation

01 Mechanical Ventilator Prototype

Embedded Hardware · Control Engineering · PCB Development · User Interface | 2020

Project Objective

Development of a functional ventilator prototype combining physical lung modelling, state-based control, electronic sensor and actuator interfacing, a manufactured PCB, and a user interface in one integrated system.

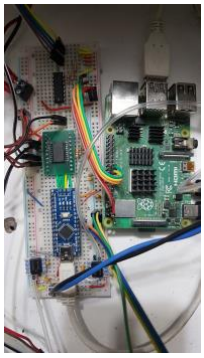
- State-based grey-box control with a fixed model structure: continuous estimation of plant parameters and state variables, with the estimates fed to an LQR controller for optimal control.
- Circuit design, simulation, and step-by-step bring-up from a protoboard to an integrated PCB.
- PCB layout, manufacturing files, and transfer to external PCB fabrication.
- Integration of sensors, actuator control, a microcontroller, and Raspberry Pi.
- C-based user interface on Raspberry Pi for displaying and setting ventilation parameters.
- Functional testing and iterative troubleshooting at both hardware and system levels.



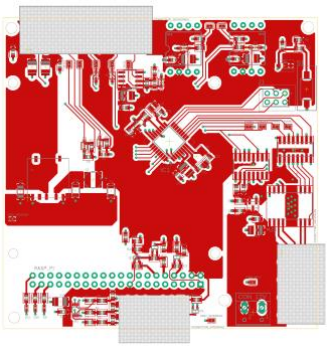
Integrated hardware prototype with Raspberry Pi, microcontroller, and custom PCB.

Production-Ready PCB

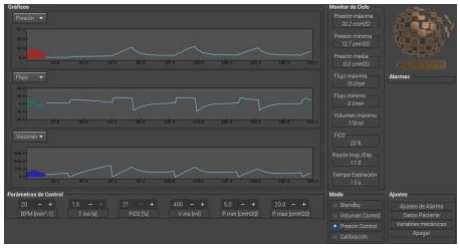
The DFM report documents a double-sided PCB with 2 copper layers, 115 nets, 249 SMD pads on the top side and 32 on the bottom side; the inspection result was reported as “QED OK.”



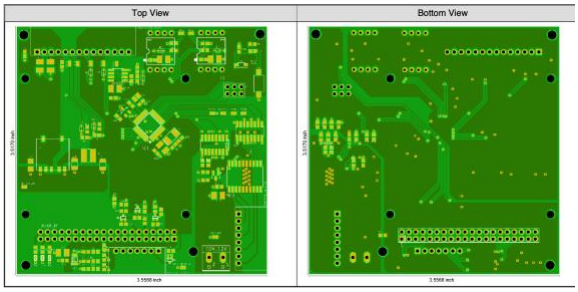
Initial Functional Prototype / Bring-Up



PCB Layout / Top Layer



User Interface on Raspberry Pi



Layout with verified DFM report

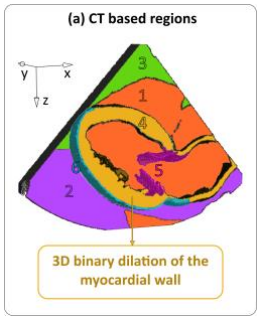
Focus: Embedded Electronics · Grey-Box Modelling · State and Parameter Estimation · LQR · PCB Design & Manufacturing · C · Raspberry Pi · Prototyping & Testing

02 Dynamic cardiac elastography simulation

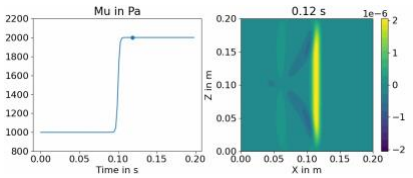
Numerical Modelling · Ultrasound Simulation · Signal Processing · In-Silico Validation | 2023–2026

Project objective: Development of an end-to-end simulation pipeline for investigating and validating cardiac time-harmonic elastography. Patient-specific 3D anatomy derived from CT data was combined with wave-mechanical finite-difference modelling in Python/Devito, ultrasound simulation in Field II, and subsequent signal processing.

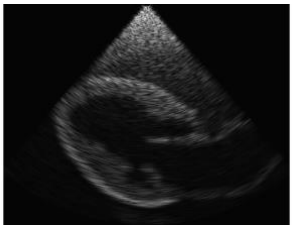
1 CT-Based 3D Geometry	2 Devito Finite-Difference Model	3 Field II Ultrasound Simulation	4 Signal Processing & Validation



CT-Based Anatomical Regions



Time-Dependent Mechanical Wave Simulation



Synthetic Ultrasound Image

Technical Contributions

- Processing and conversion of CT-based 3D anatomies into simulation-ready regions and material models.
- Finite-difference modelling of time-dependent mechanical response and wave propagation in Python using Devito.
- Coupling of the mechanical simulation with Field II to generate realistic synthetic ultrasound data.
- Signal processing and quantitative analysis of simulated measurement data for methodological validation.

Results & Impact

The simulation pipeline combined anatomical modelling, computational physics, and ultrasound signal processing into a reproducible validation framework. The results contributed to a peer-reviewed publication.

Publication

Klemmer Chandía, S. et al. “Cardiac Multifrequency Time-harmonic Elastography: In Silico Validation and In Vivo Application.” Ultraschall in der Medizin (2026). DOI: 10.1055/a-2852-6188.

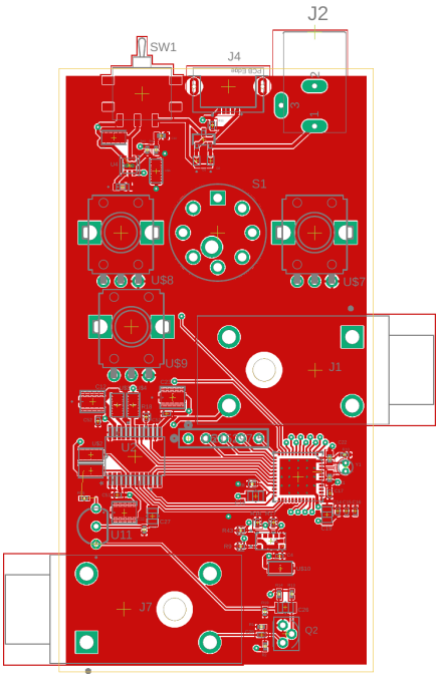
Focus: 3D CT · Devito · Finite Differences · Field II · Ultrasound Physics · Signal Processing · Numerical Validation · Python/MATLAB

03 Auto-Wah guitar pedal

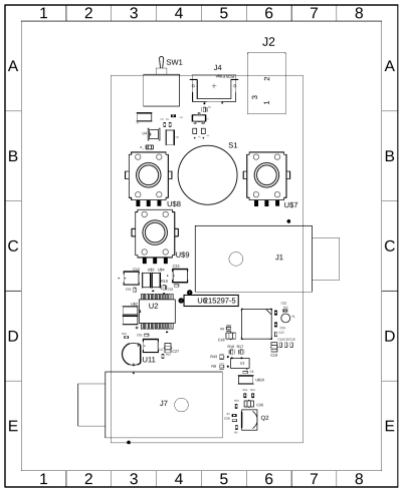
Mixed-signal electronics · Circuit design · PCB-Design | University project, 2020

Project objective: Development of an auto-wah guitar effects pedal, covering the electronic circuit, simulation and component selection through to a complete, technically reviewed PCB layout.

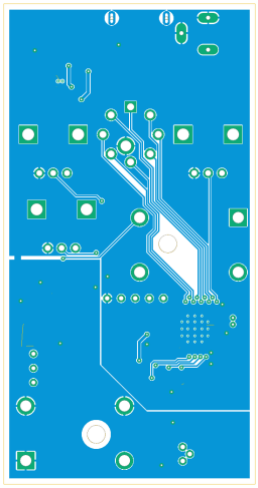
- Design of the audio circuit including analog input and output stages and digital two-way conversion.
- Circuit simulation prior to PCB layout.
- Schematic capture, component placement, PCB layout in EAGLE and sizing of its components.
- Integration of controls, audio connectors, and power connections.
- Routing of signal and power nets, including ground planes; final design review and approval by an experienced PCB designer.



PCB Top Layer and Component Placement in EAGLE.



Component and Enclosure Planning



PCB Bottom Layer / Routing

Technical Focus

The project demonstrates practical experience across the ECAD workflow through to design approval: circuit design and simulation, schematic capture, placement, routing, and mechanical integration. The PCB design was technically reviewed and approved but was not manufactured.

Focus: EAGLE · LTspice · Analog Circuit Design · Audio Signal Processing · PCB Layout · Design Review